IMPERIAL

Department of Chemical Engineering

Machine Learning for Chemical Engineering

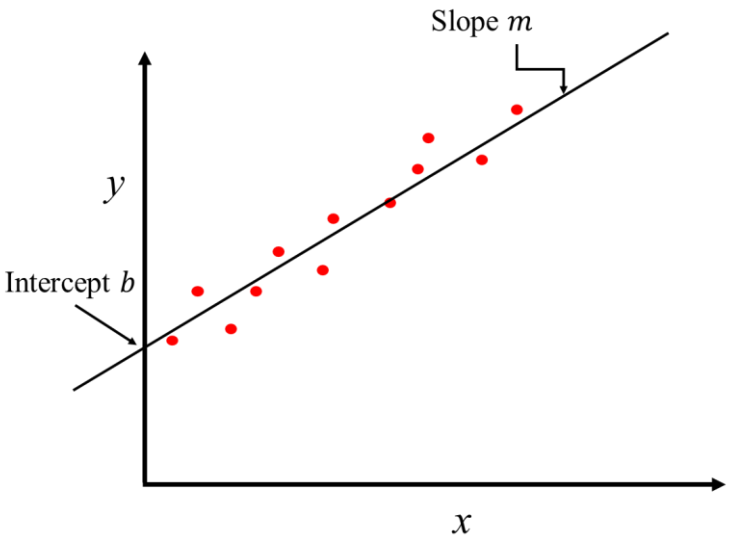
Lecture 1 – Maximum Likelihood Estimation (MLE)

Maximum Likelihood Estimation (MLE)

Why are we using least-square?

$$\min_{\boldsymbol{\theta}} \sum_{i=1}^{n_d} \left(\boldsymbol{\theta}^T \mathbf{x}^{(i)} - y^{(i)}\right)^2$$

What assumptions are we making?



Maximum Likelihood Estimation (MLE) is a <u>statistical method</u> used to estimate the parameters of a model based on observed data.

Maximum Likelihood Estimation (MLE)

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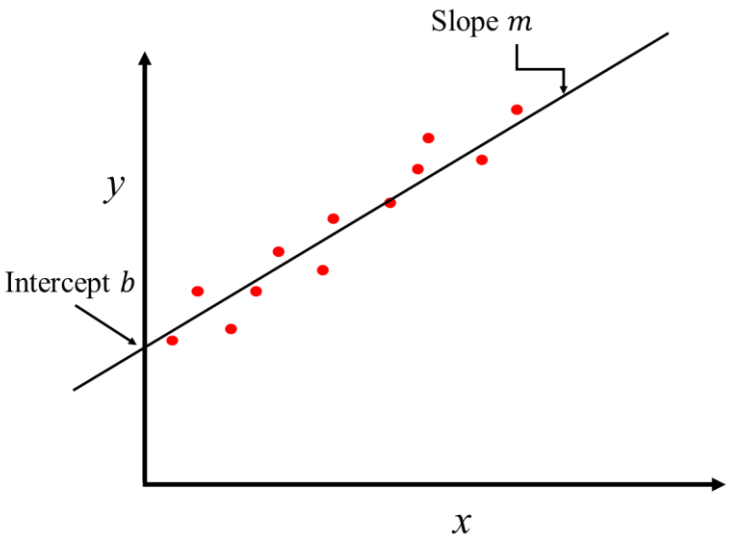
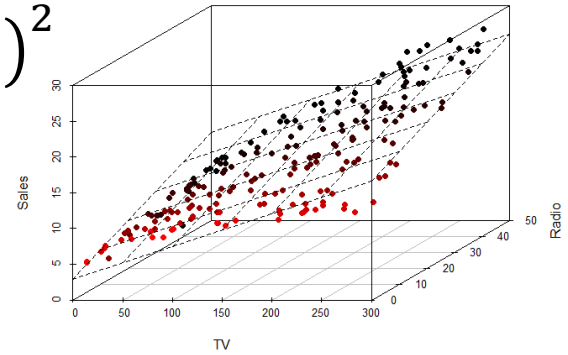


What assumptions are we making?

Remember:

$$\min_{\boldsymbol{\theta}} \sum_{i=1}^{n_d} \left(\theta_0 + \theta_1 x_1^{(i)} + \theta_2 x_2^{(i)} + \cdots + \theta_{n_x} x_{n_x}^{(i)} - y^{(i)}\right)^2 = \min_{\boldsymbol{\theta}} \sum_{i=1}^{n_d} \left(\boldsymbol{\theta}^T \mathbf{x} - y^{(i)}\right)^2$$

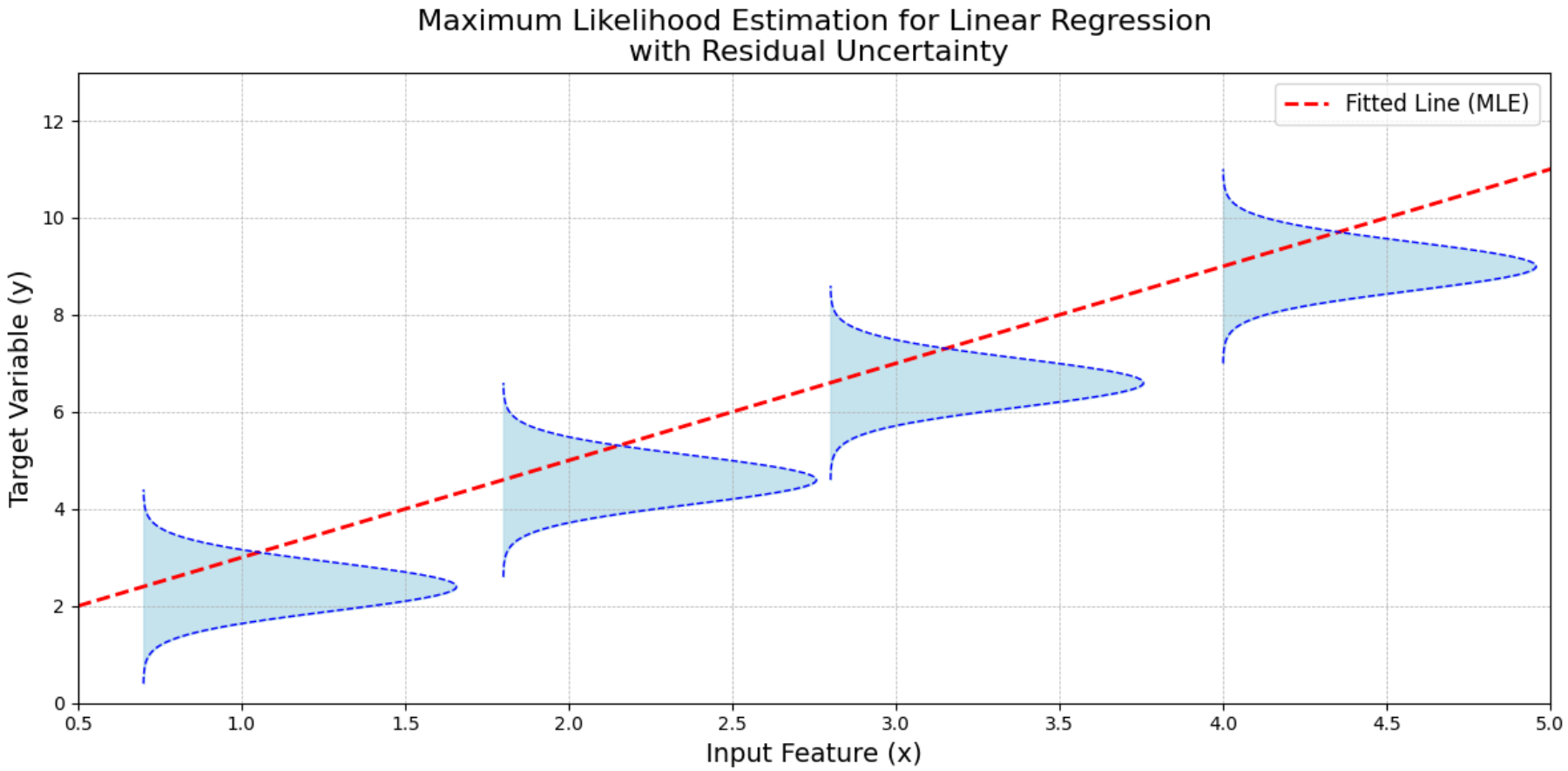
$$\boldsymbol{\theta}^T \mathbf{x} = \theta_0 + \theta_1 x_1^{(i)} + \theta_2 x_2^{(i)} + \cdots + \theta_{n_x} x_{n_x}^{(i)}$$



Maximum Likelihood Estimation (MLE)

Given a system from which we get measurements we can assume:

$$y = f(\mathbf{x}) + \varepsilon \qquad \text{with} \quad \varepsilon \sim \mathcal{N}(0, \sigma^2)$$



If we generate samples, they will be drawn from Gaussian distributions

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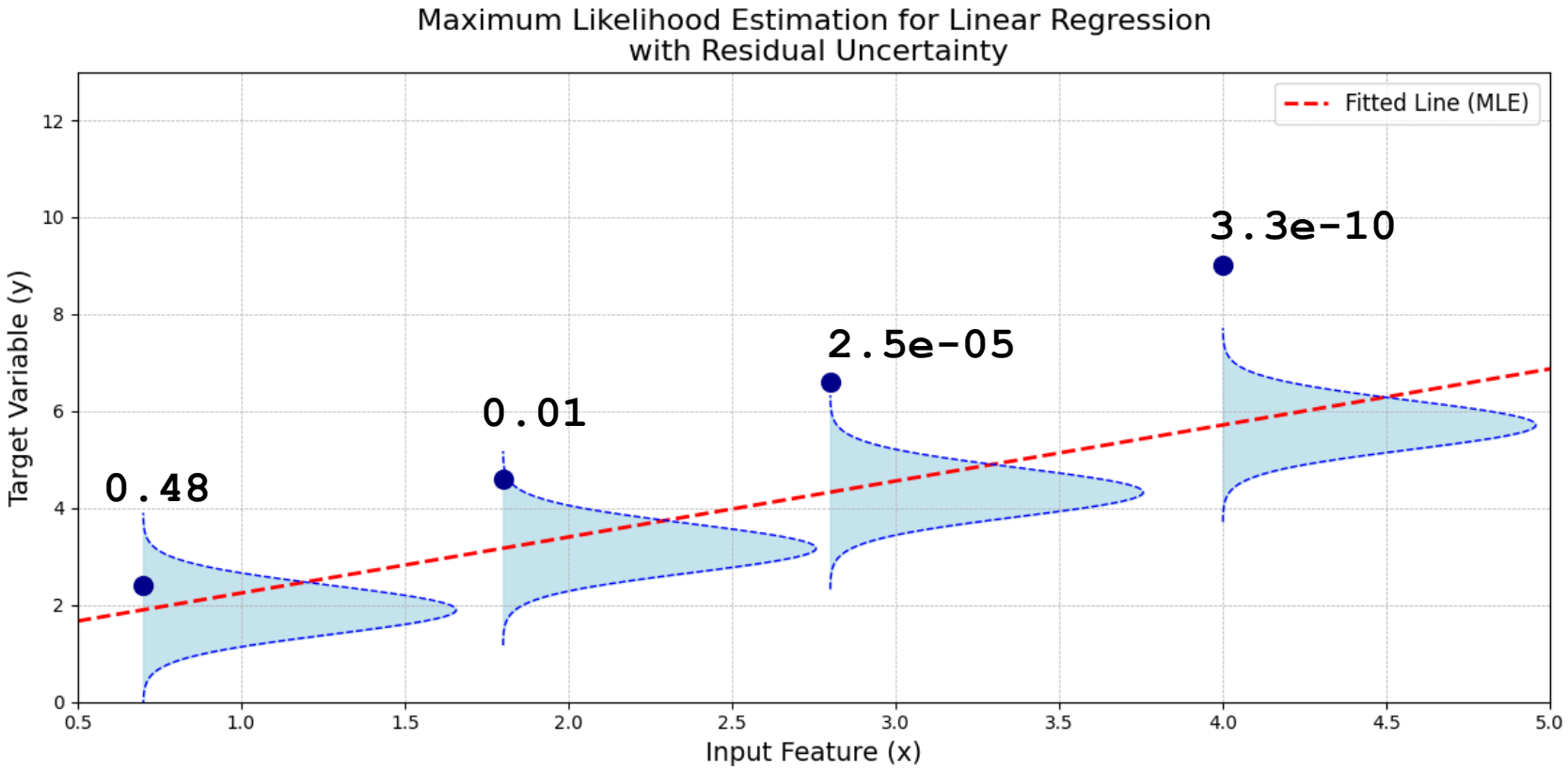
Normally distributed residuals:

$$p(y_i | x, \theta, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{\left(y^{(i)} - mx^{(i)} + b\right)^2}{2\sigma^2}}$$



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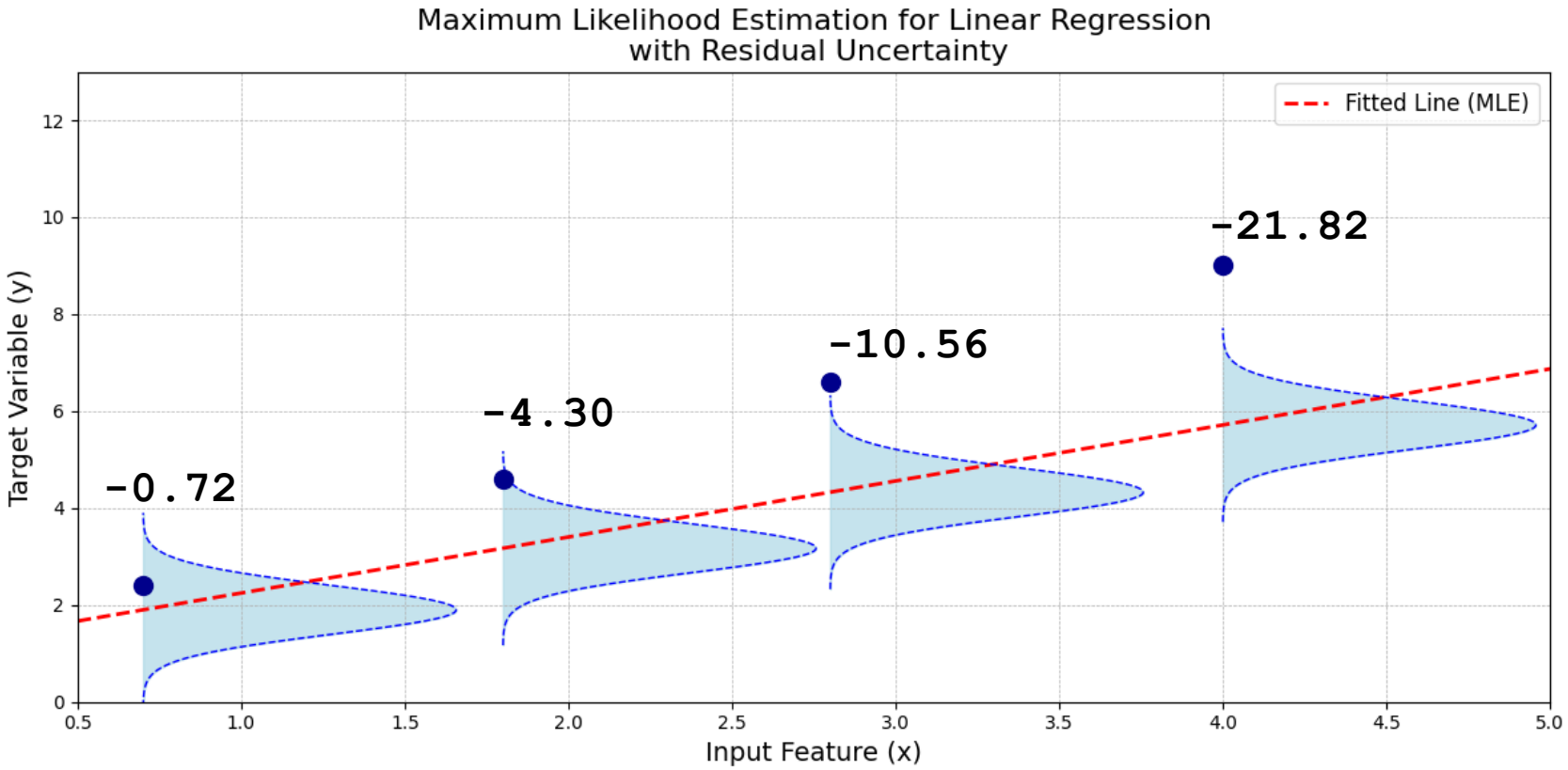

Taking the natural log:

$$\ln(p) = -\ln(2\pi\sigma^2) - \frac{\left(y^{(i)} - mx^{(i)} + b\right)^2}{2\sigma^2}$$

Sum of values:
-37.41

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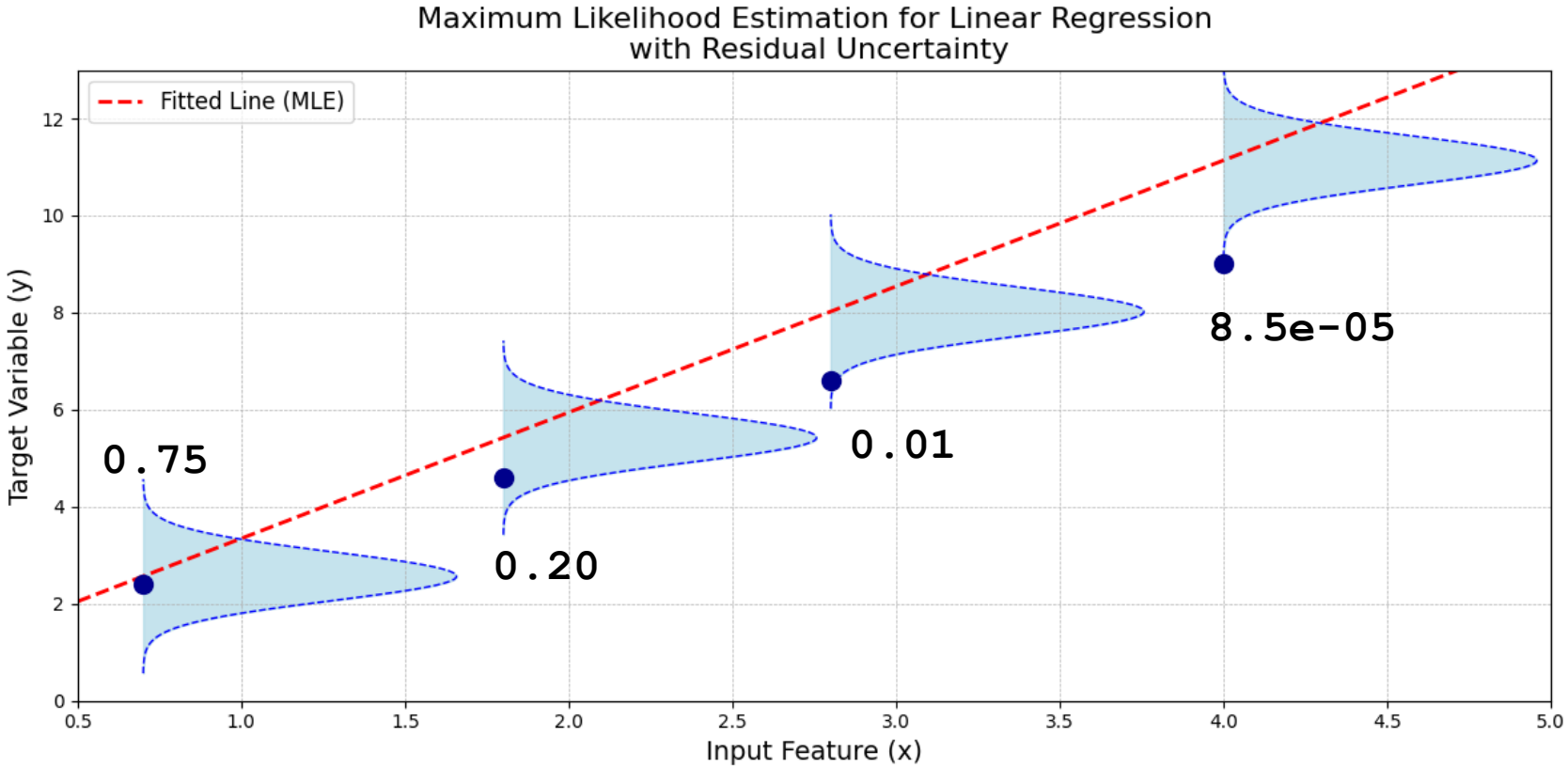

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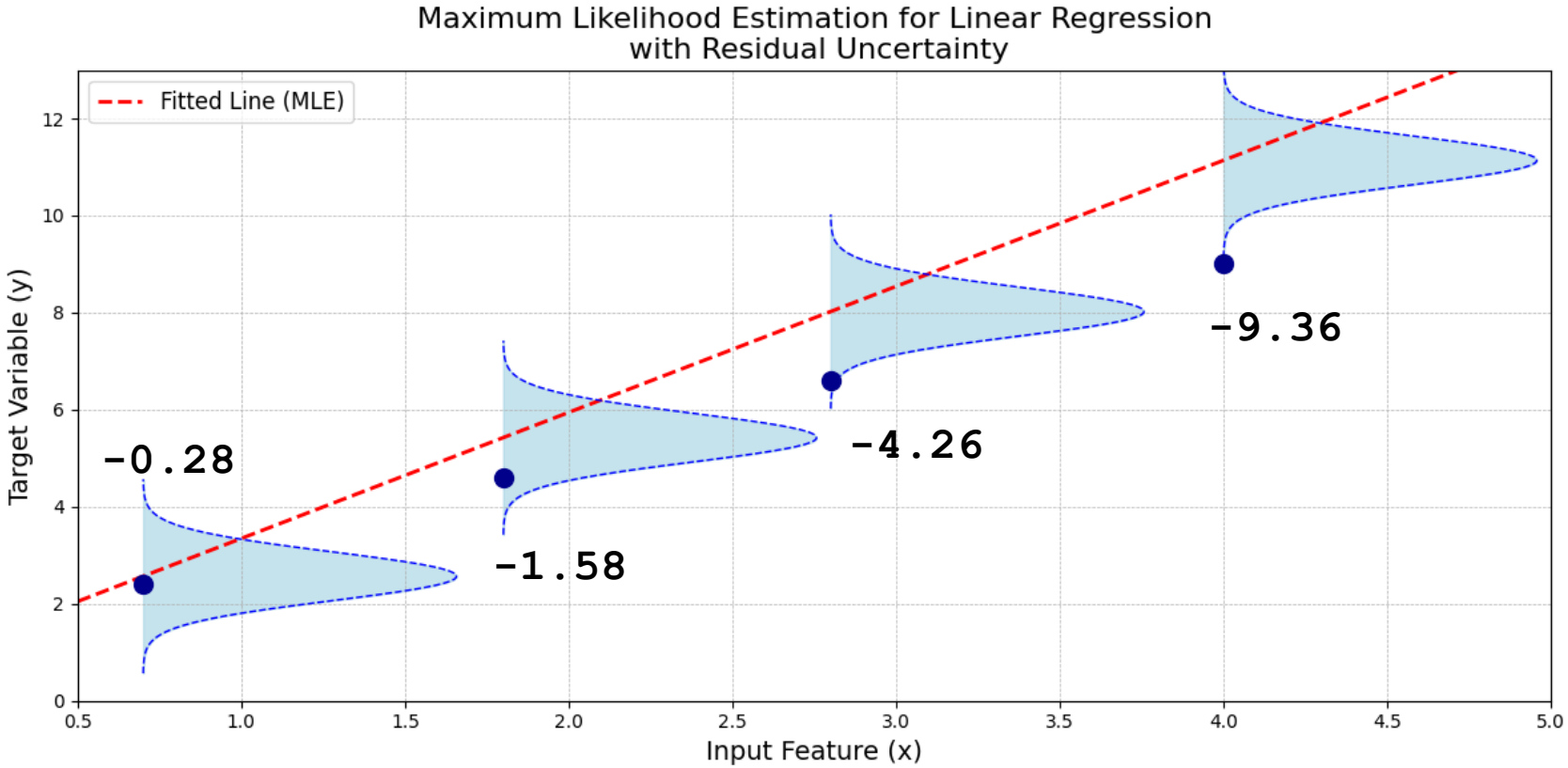

Taking the natural log:

$$\ln(p) = -\ln(2\pi\sigma^2) - \frac{\left(y^{(i)} - mx^{(i)} + b\right)^2}{2\sigma^2}$$

Sum of values:
-15.49

Maximum Likelihood Estimation (MLE)

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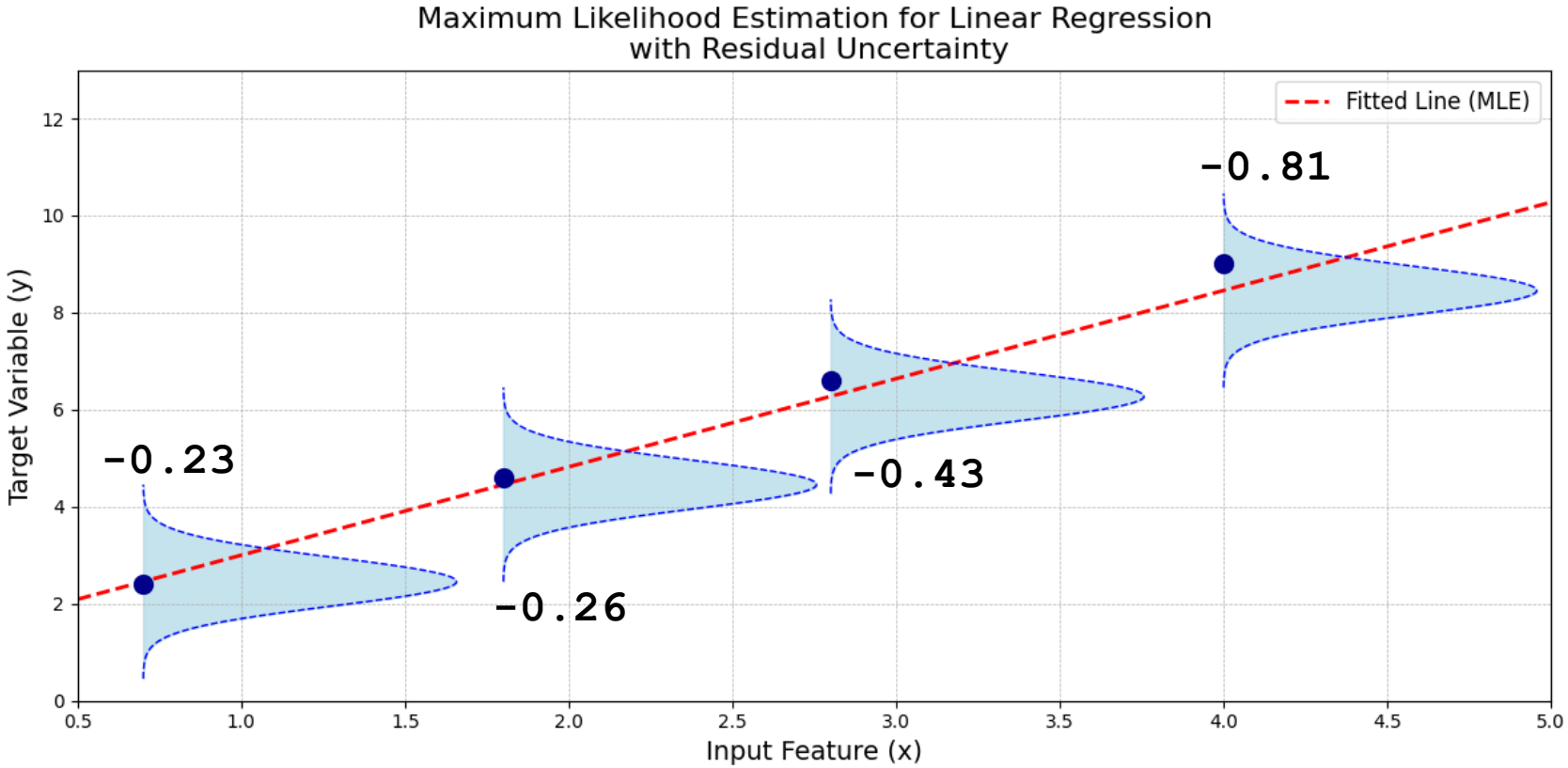

Taking the natural log:

$$\ln(p) = -\ln(2\pi\sigma^2) - \frac{\left(y^{(i)} - mx^{(i)} + b\right)^2}{2\sigma^2}$$

Sum of values:
-1.75

Maximum Likelihood Estimation (MLE)

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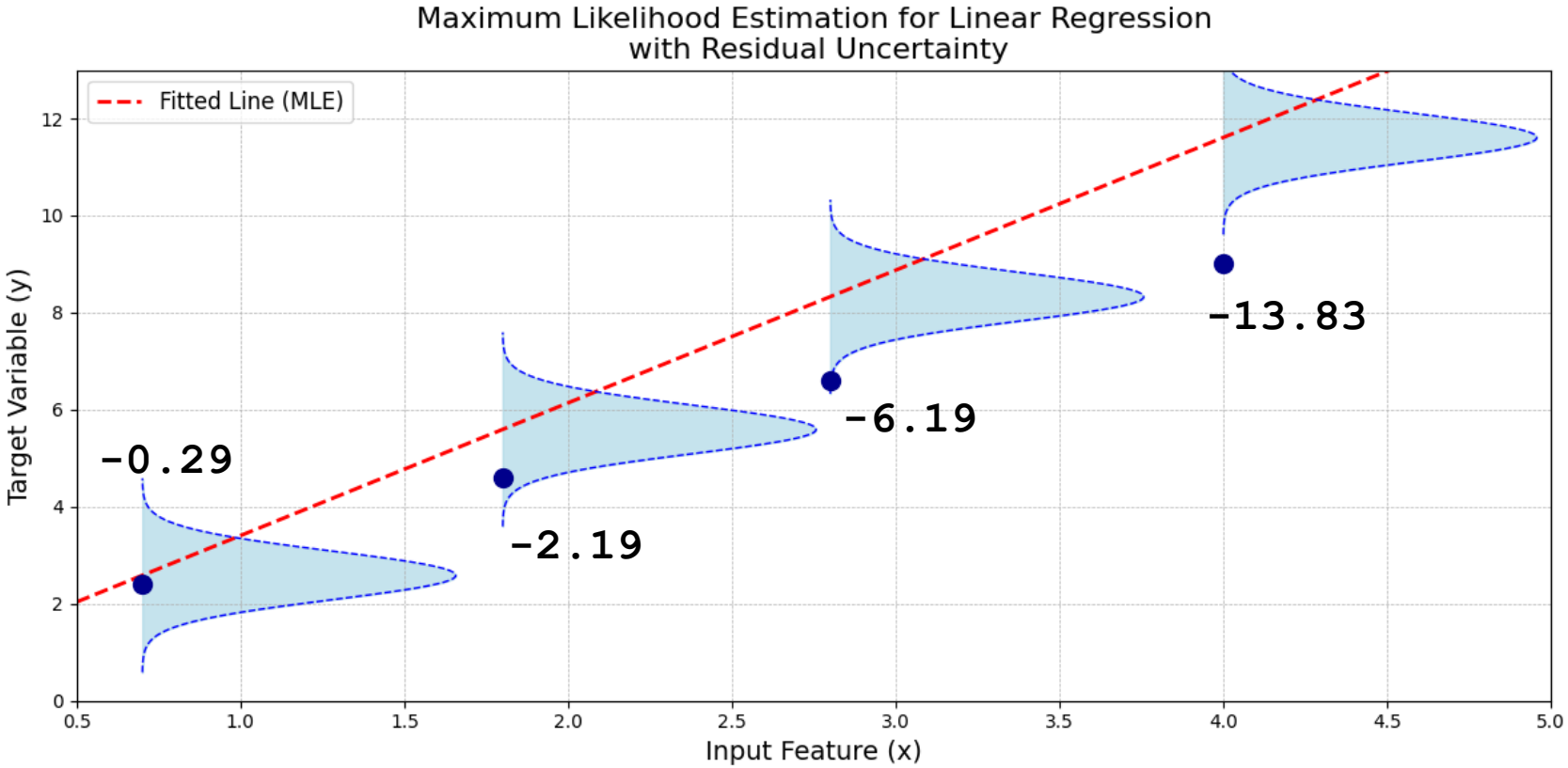

Taking the natural log:

$$\ln(p) = -\ln(2\pi\sigma^2) - \frac{\left(y^{(i)} - mx^{(i)} + b\right)^2}{2\sigma^2}$$

Sum of values:

-22.51

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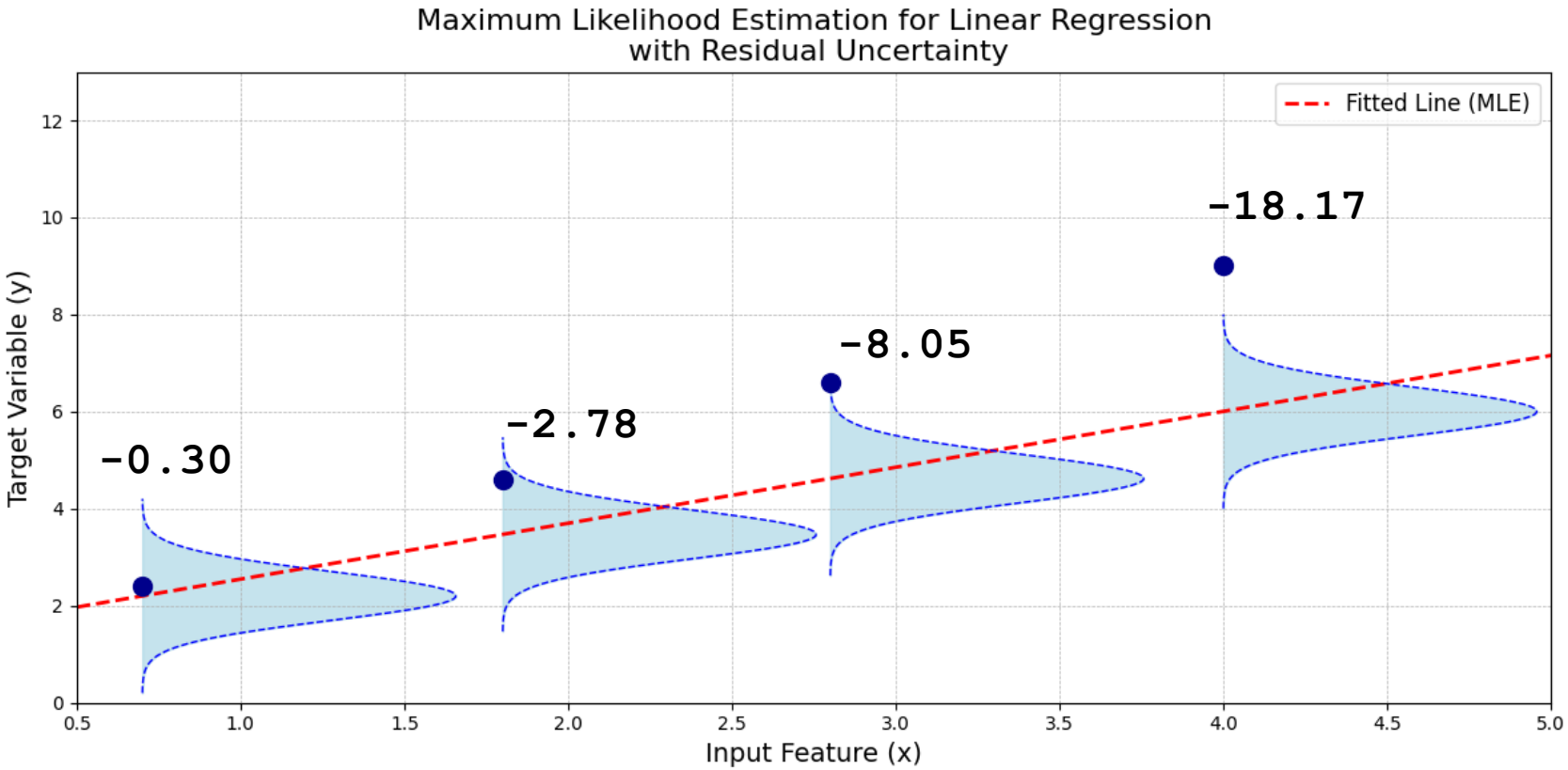


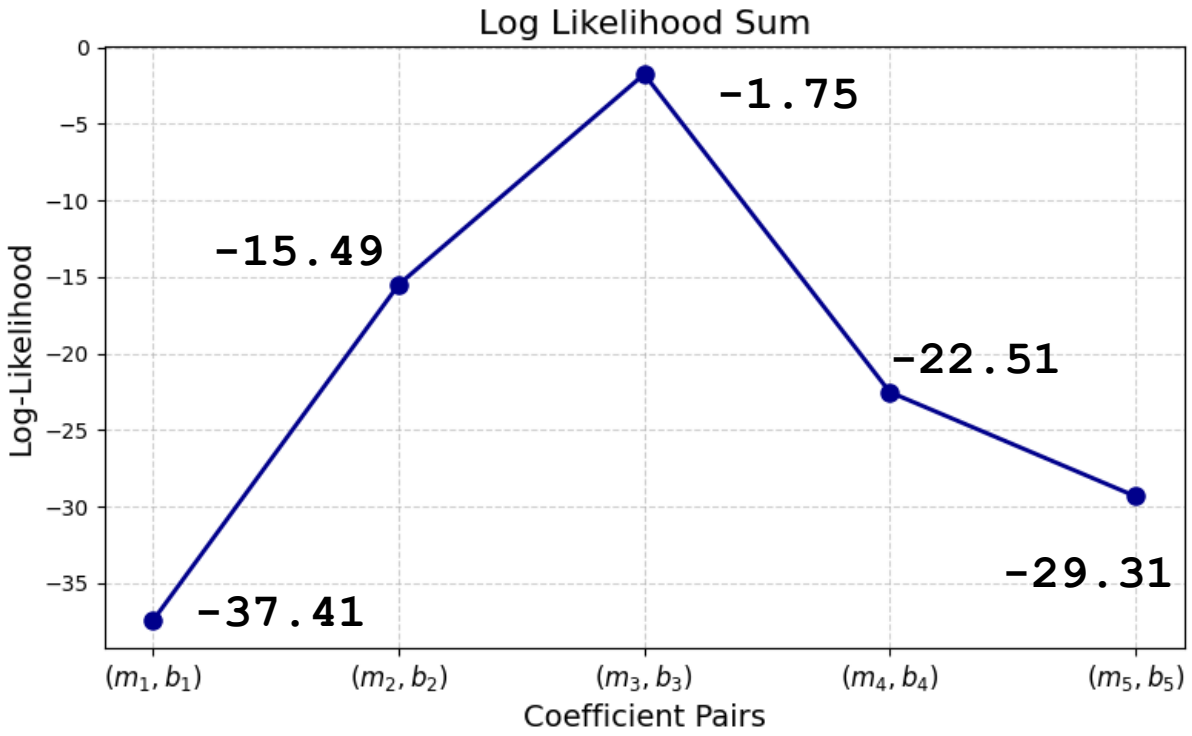
Taking the natural log:

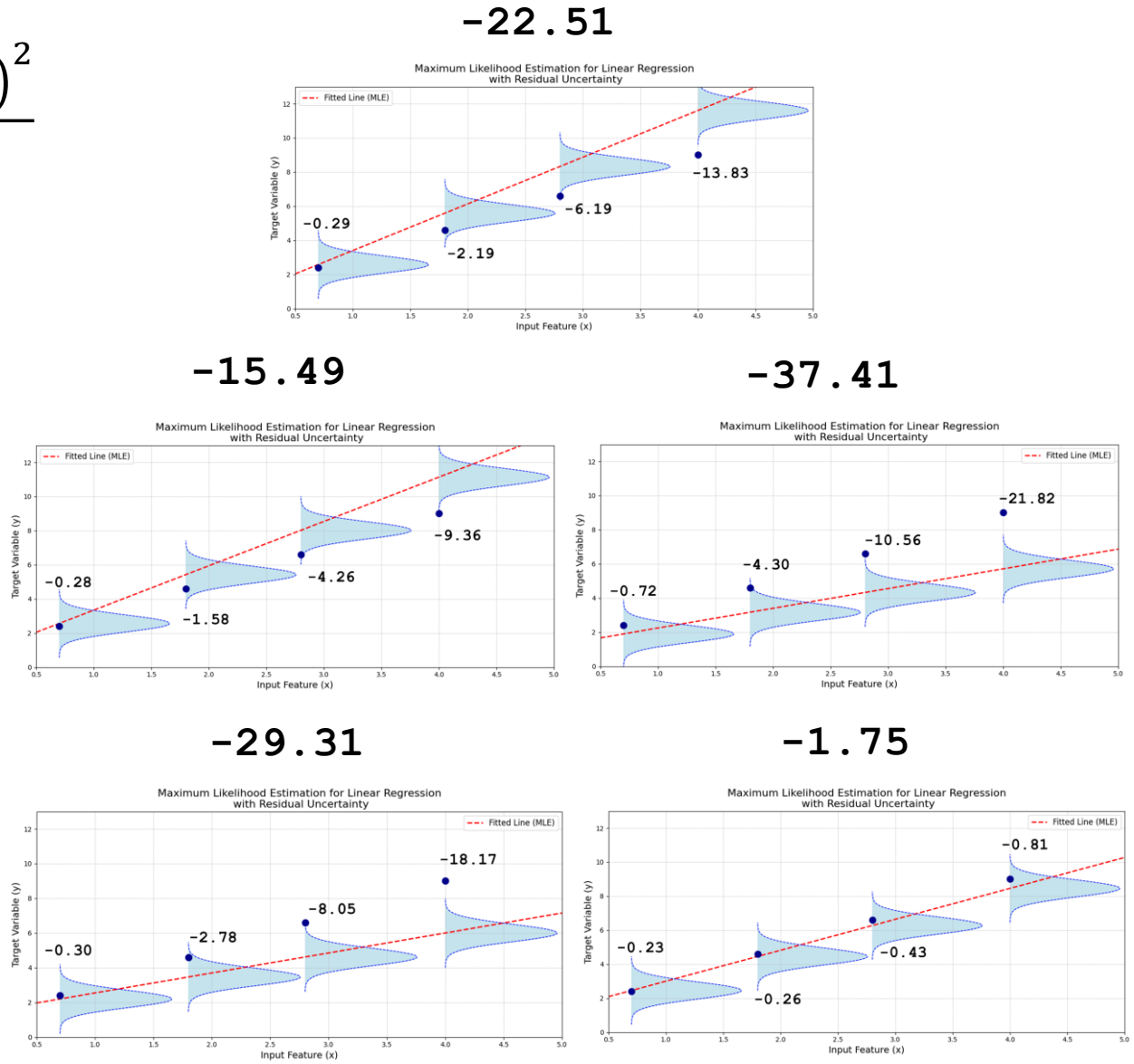
$$\ln(p) = -\ln(2\pi\sigma^2) - \frac{\left(y^{(i)} - mx^{(i)} + b\right)^2}{2\sigma^2}$$

Sum of values:
-29.31

Maximum Likelihood Estimation (MLE)

$$\text{log likelihood} = -4\ln(2\pi\sigma^2) - \sum_{i=1}^{4} \frac{\left(y^{(i)} - mx^{(i)} + b\right)^2}{2\sigma^2}$$

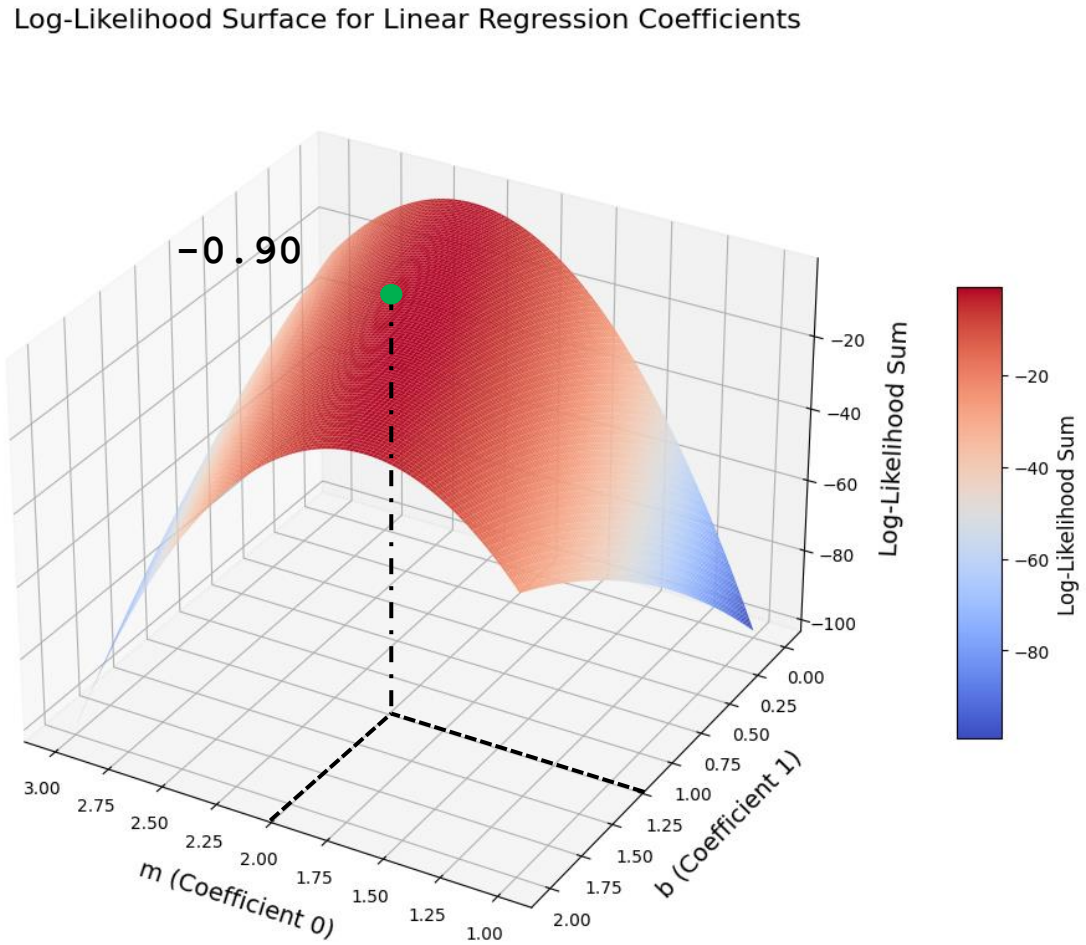


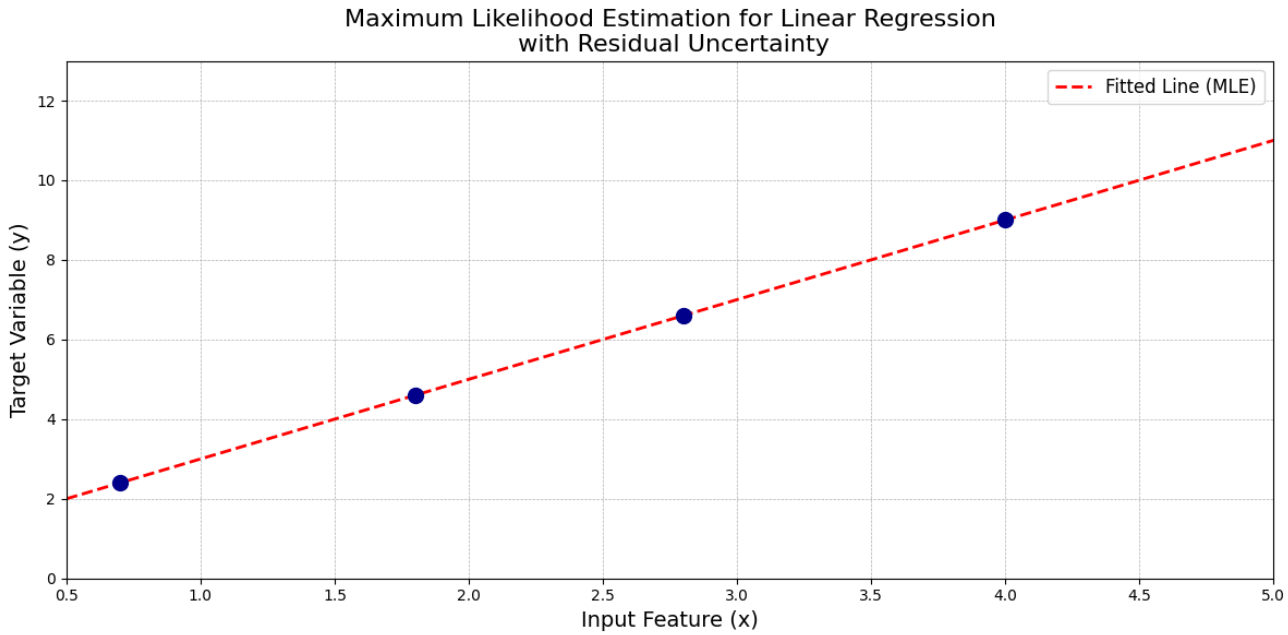


Maximum Likelihood Estimation (MLE)

$$\text{log likelihood} = -4\ln(2\pi\sigma^2) - \sum_{i=1}^{4} \frac{\left(y^{(i)} - mx^{(i)} + b\right)^2}{2\sigma^2}$$

$$\max_{m,b} - \sum_{i=1}^{4} \frac{\left(y^{(i)} - mx^{(i)} + b\right)^2}{2\sigma^2}$$





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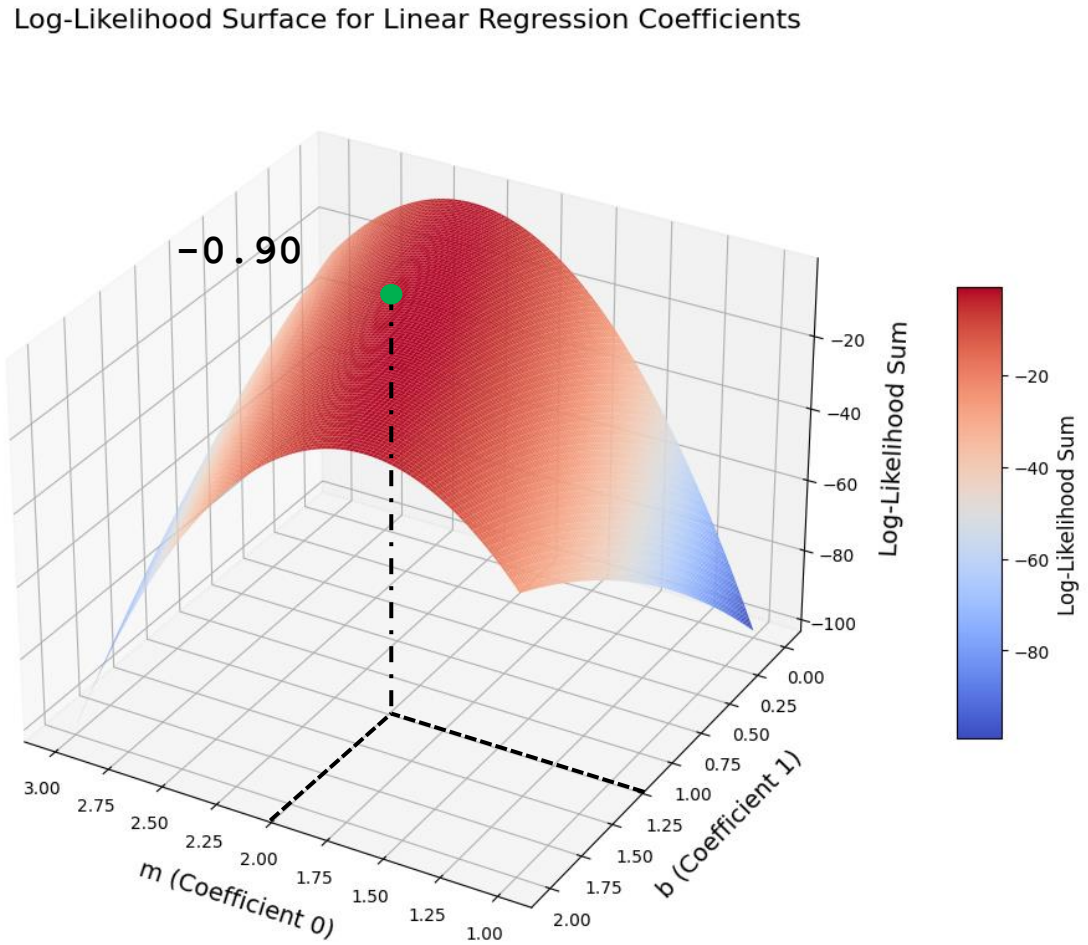

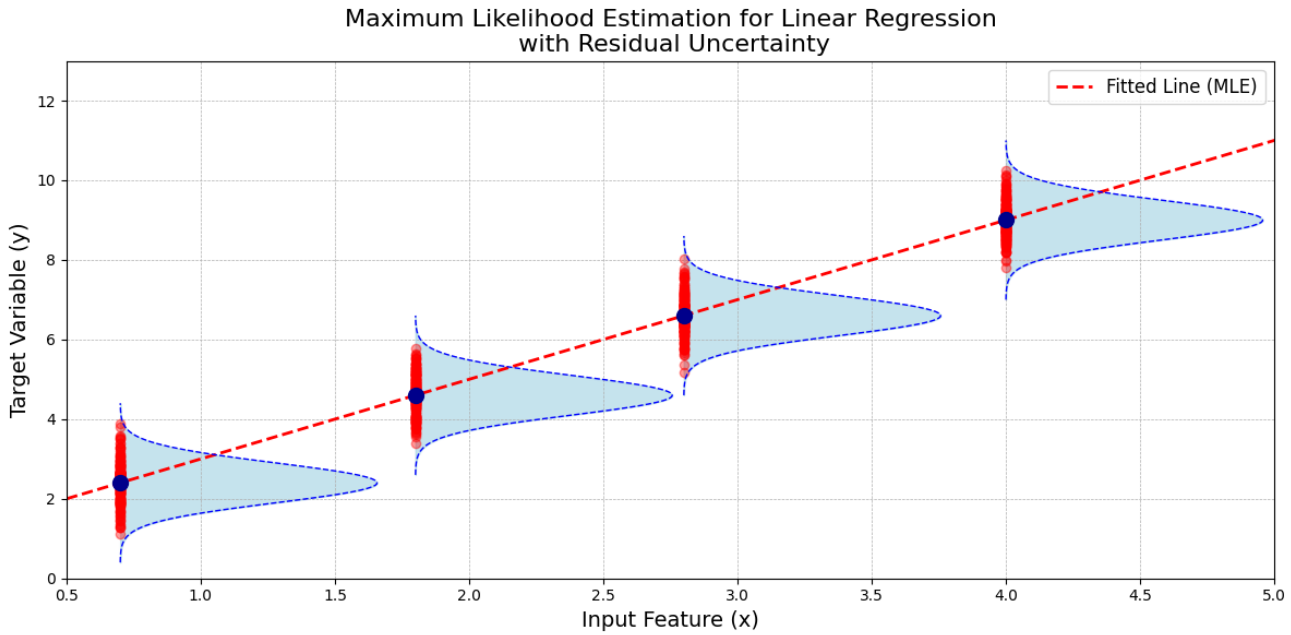

Maximum Likelihood Estimation (MLE) - Summary

Given a system from which we get measurements we can assume:

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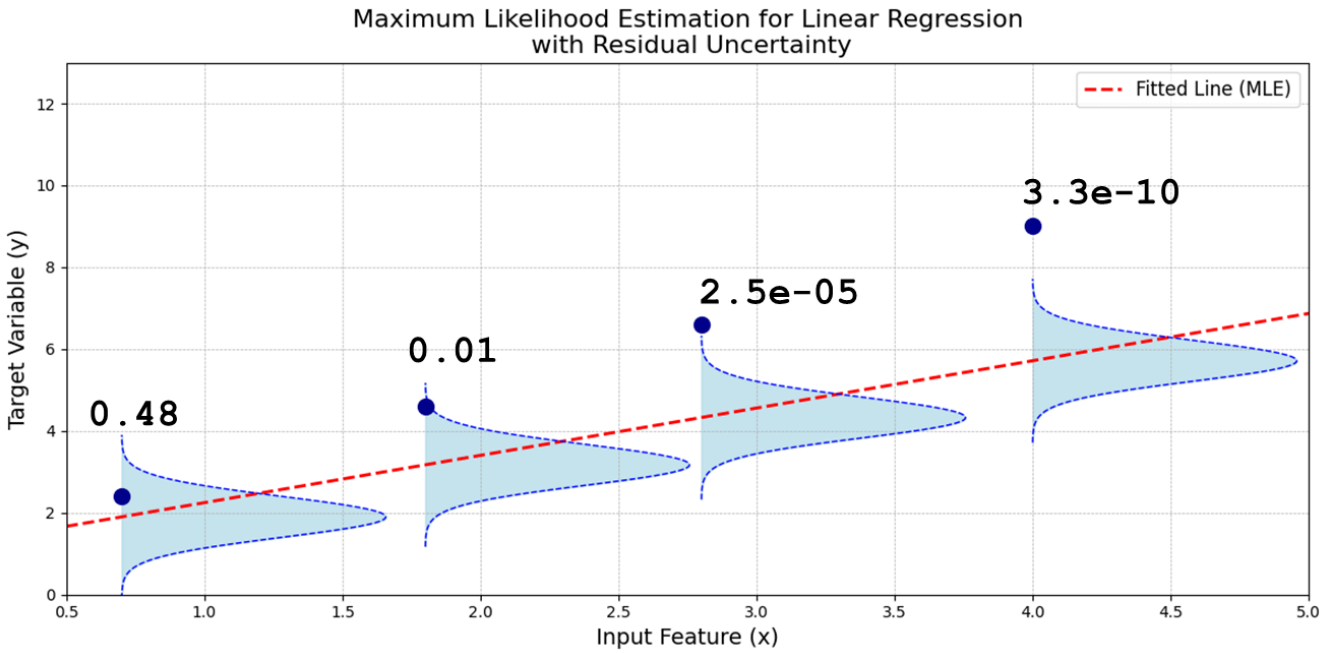
Likelihood of n_d independent datapoints produced by $\hat{f}(\mathbf{x}) \coloneqq \boldsymbol{\theta}^T \mathbf{x}$:

$$p(y|\mathbf{x}, \boldsymbol{\theta}, \sigma^2) = \prod_{i=1}^{n_d} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{\left(y^{(i)} - \boldsymbol{\theta}^T \mathbf{x}^{(i)}\right)^2}{2\sigma^2}}$$



We wish to maximize the likelihood of the data being produced by the model

$$\hat{f}(\mathbf{x}) \coloneqq \boldsymbol{\theta}^T \mathbf{x}$$

which yields:

$$\max_{\boldsymbol{\theta}} \prod_{i=1}^{n_d} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{\left(y^{(i)} - \boldsymbol{\theta}^T \mathbf{x}^{(i)}\right)^2}{2\sigma^2}}$$

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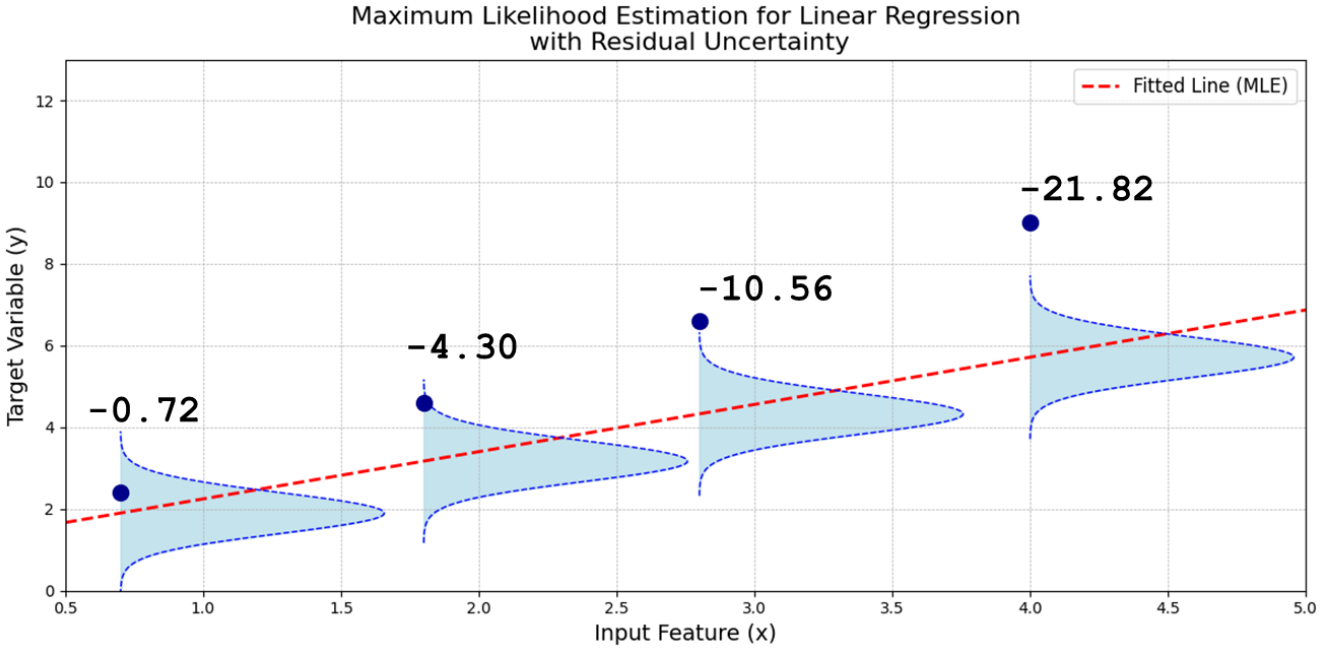

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taking the logarithm:

$$\max_{\boldsymbol{\theta}} -\frac{n_d}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^{n_d} \left(y^{(i)} - \boldsymbol{\theta}^T \mathbf{x}^{(i)}\right)^2$$

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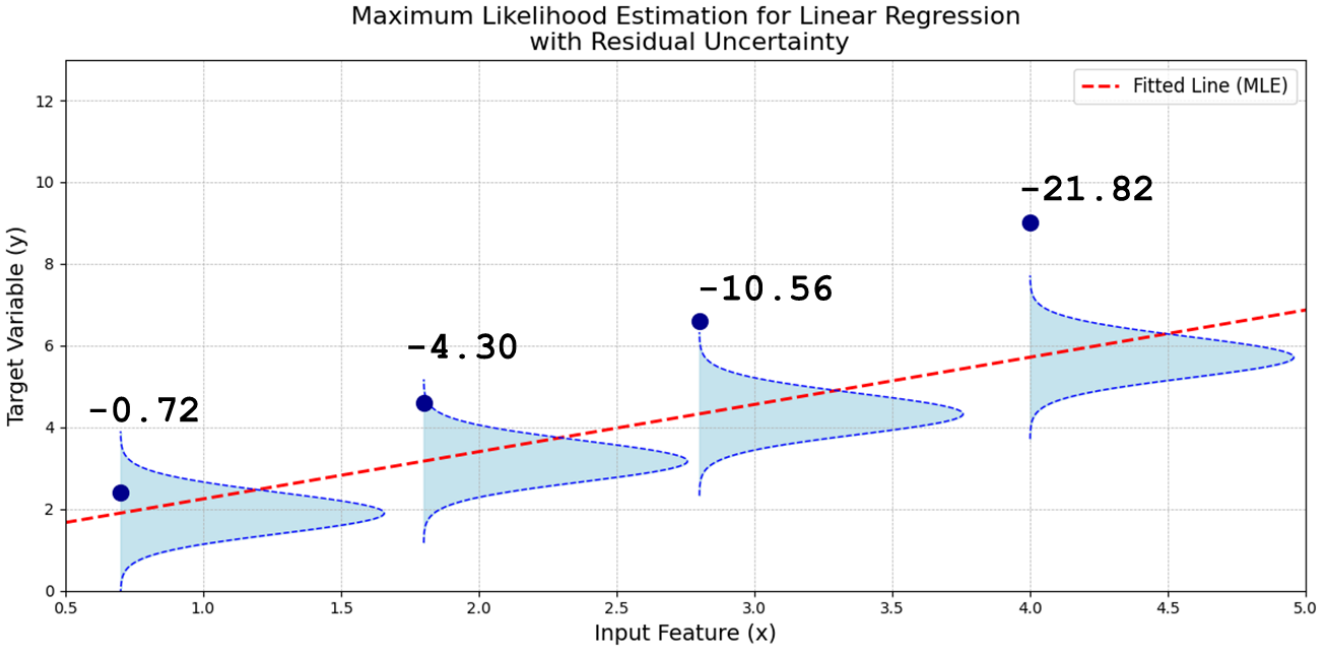

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$$\hat{f}(\mathbf{x}) \coloneqq \boldsymbol{\theta}^T \mathbf{x}$$

Minimizing and noting σ is constant:

<u>Least squares!</u>

$$\min_{\boldsymbol{\theta}} \frac{1}{2\sigma^2} \sum_{i=1}^{n_d} \left(y^{(i)} - \boldsymbol{\theta}^T \mathbf{x}^{(i)}\right)^2$$